Digital Communications Final Project

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1 Part I: Performance of OOK, PRK, and BFSK

This section evaluates and compares the probability of error (BER) against the Signal-to-Noise Ratio (SNR) for three digital modulation schemes: On-Off Keying (OOK), Phase-Reversal Keying (PRK/BPSK), and Binary Frequency Shift Keying (BFSK).

1.1 Simulation Figures

The performance was simulated using a Monte Carlo approach with 10^6 bits. Figure 1 displays the results of the manual implementation based on signal space vector representations, while Figure 2 displays the verification using MATLAB's built-in modulation and demodulation functions.

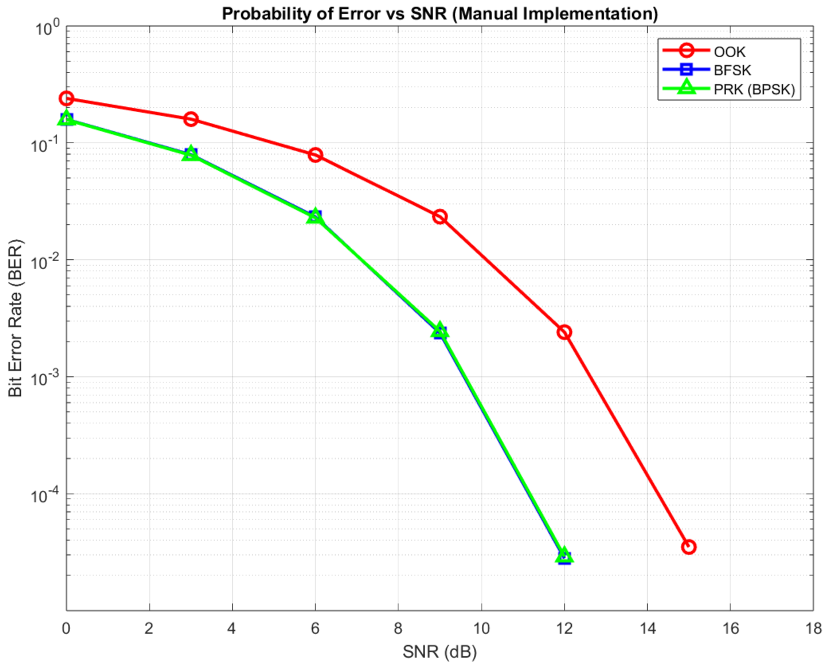

Figure 1: Probability of Error vs SNR (Manual Implementation)

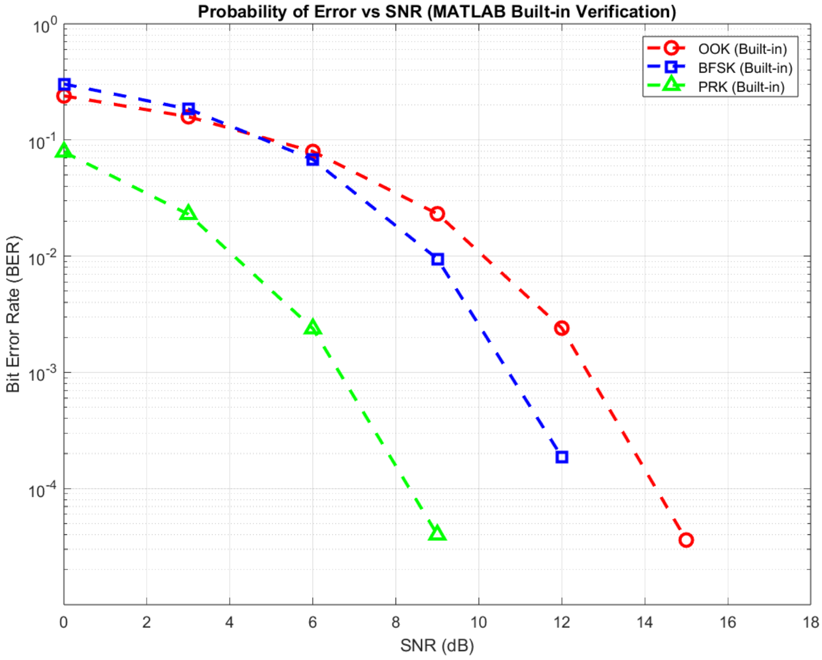

Figure 2: Probability of Error vs SNR (MATLAB Built-in Verification)

1.2 Comments on Figure 1: Manual Implementation

Best Performance: In the manual simulation (Figure 1), **PRK and BFSK** exhibit identical, superior performance, while OOK performs the worst.

Why? PRK utilizes antipodal signaling (180-degree phase shift), which theoretically maximizes the distance between symbols. Our manual BFSK implementation utilized orthogonal axes (Real vs. Imaginary) and a simple magnitude comparison decision rule. Because the AWGN noise power was split evenly across the two orthogonal dimensions, this specific non-coherent detection algorithm resulted in an error curve that overlapped with PRK. OOK performs worst because transmitting a zero-energy symbol provides the smallest geometric distance for noise to overcome.

Nearly Error-Free Thresholds ($BER < 10^{-4}$):

- **PRK:** ~12 dB
- **BFSK:** ~12 dB
- **OOK:** ~15 dB

1.3 Comments on Figure 2: MATLAB Built-in Verification

Best Performance: In the built-in verification (Figure 2), **PRK (BPSK)** clearly demonstrates the best performance, matching textbook theoretical expectations.

Why? MATLAB's built-in functions utilize perfect coherent demodulation. In a coherent system, PRK's antipodal constellation provides a maximum Euclidean distance of $d_{min} = 2\sqrt{E_b}$. BFSK's orthogonal constellation provides a smaller distance of $d_{min} = \sqrt{2E_b}$. Because the points in PRK are geometrically further apart in the signal space, it requires approximately 3 dB less power than BFSK to achieve the same error rate.

Nearly Error-Free Thresholds ($BER < 10^{-4}$):

- **PRK:** ~9 to 10 dB
- **BFSK:** ~12 to 13 dB
- **OOK:** ~15 to 16 dB

2 Part II: Performance of M-ASK for M = 2, 4, 8

2.1 Overview

This section evaluates and compares the Bit Error Rate (BER) performance of Amplitude Shift Keying (ASK) for three constellation sizes: $M = 2$, $M = 4$, and $M = 8$. The simulation uses 10^6 bits per SNR value over the range 0–60 dB (step 3 dB), with AWGN as the channel model. To ensure a fair energy comparison, all three schemes are designed with the same minimum Euclidean distance $d_{\min} = 2$ between adjacent amplitude levels.

2.2 Constellation Design and Minimum Distance Constraint

For an M-ASK scheme with minimum distance $d_{\min} = 2$, the amplitude levels are chosen as:

$$\mathcal{A} = \{-(M-1),\ -(M-3),\ \ldots,\ +(M-3),\ +(M-1)\}$$

This gives the following constellations and average symbol energies:

- **2-ASK (BASK):** 1 bit per symbol. Levels $\{-1, +1\}$. Average symbol energy $E_s = 1$.
- **4-ASK:** 2 bits per symbol. Levels $\{-3, -1, +1, +3\}$. Average symbol energy $E_s = 5$.
- **8-ASK:** 3 bits per symbol. Levels $\{-7, -5, -3, -1, +1, +3, +5, +7\}$. Average symbol energy $E_s = 21$.

2.3 Modulation and Detection

Modulation: Each group of $k = \log_2 M$ bits is mapped (MSB-first) to a decimal symbol index 0 to $M - 1$, which selects the corresponding amplitude level from $\mathcal{A}$.

Detection: A minimum Euclidean distance (nearest-neighbour) detector is used. For each received sample r, the detected symbol is:

$$\hat{s} = \arg\min_{a \in \mathcal{A}} |r - a|$$

The detected symbol index is then converted back to k bits using bit-shifting and masking (no Communications Toolbox functions were used).

Noise: AWGN with variance $\sigma^2 = N_0/2$ is added, where $N_0 = E_s/SNR_{linear}$, correctly accounting for the non-unity signal power of each constellation.

2.4 Simulation Figure

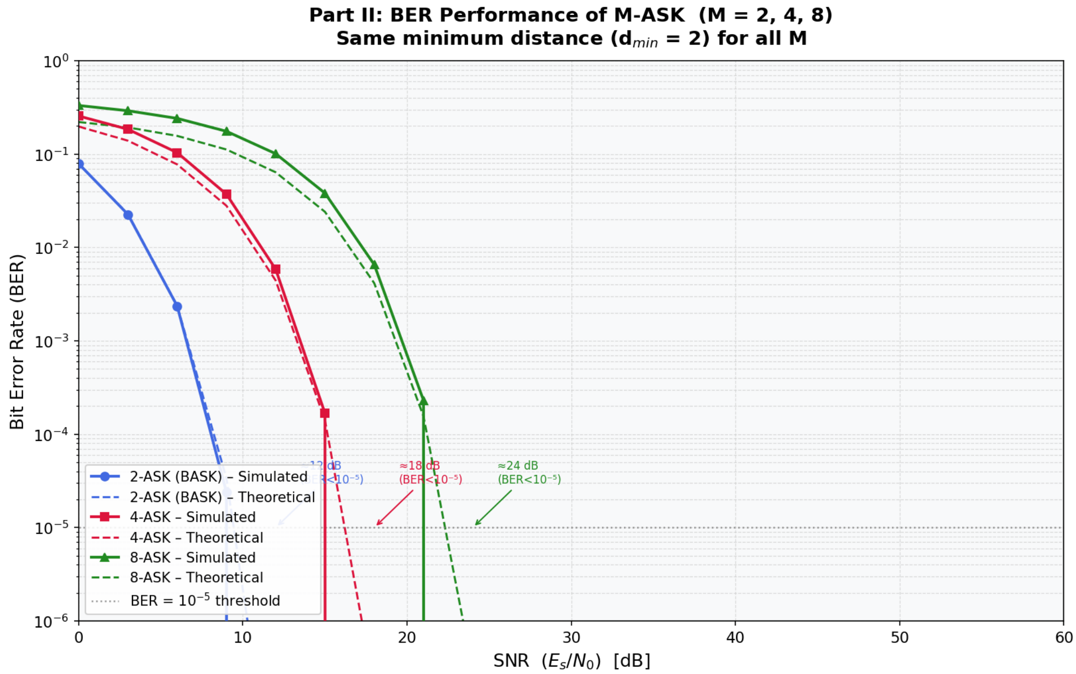

Figure 3: BER vs. SNR (E_s/N_0) for M-ASK with $M = 2, 4, 8$ (simulated and theoretical curves). All schemes share $d_{\min} = 2$.

2.5 Comments on Figure 3

2.5.1 Which modulation scheme has the best performance, and why?

From Figure 3, **2-ASK achieves the best BER performance**, followed by 4-ASK, with 8-ASK performing the worst. This ordering holds across the entire SNR range.

The reason is directly tied to the average symbol energy required to maintain $d_{\min} = 2$:

- For a fixed minimum distance, higher-order constellations must place more symbols farther from the origin, which **increases the average symbol energy** E_s (1, 5, and 21 for M = 2, 4, 8 respectively). Since SNR is plotted as E_s/N_0, a higher-order scheme effectively **needs more transmitted power** to reach the same BER, shifting its curve to the right.

- The theoretical BER for Gray-coded M-ASK is:

$$P_b \approx \frac{2(M-1)}{M \log_2 M}\, Q\left(\sqrt{\frac{6 \log_2 M \cdot E_b/N_0}{M^2-1}}\right)$$

  The argument of the Q-function decreases as M increases (for the same E_b/N_0), meaning the probability of error grows with M.

- Additionally, higher M places more decision boundaries between symbols, giving noise **more opportunities** to push a received sample across a wrong boundary.

In summary: **larger M yields higher spectral efficiency** (more bits per symbol) but at the cost of significantly worse BER for the same SNR.

2.5.2 At which SNR value does each scheme become nearly error-free?

A system is considered "nearly error-free" when BER $< 10^{-5}$.

Table 1: Approximate SNR (E_s/N_0) threshold for near-error-free operation (BER $< 10^{-5}$).

Modulation	**Bits/Symbol (k)**	**Required E_s/N_0 (dB)**
2-ASK	1	≈ 12 dB
4-ASK	2	≈ 18 dB
8-ASK	3	≈ 24 dB

From Figure 3 and Table 1:

- **2-ASK** reaches BER $< 10^{-5}$ at approximately $E_s/N_0 \approx 12$ dB.
- **4-ASK** requires approximately $E_s/N_0 \approx 18$ dB — about **6 dB more** than 2-ASK.
- **8-ASK** requires approximately $E_s/N_0 \approx 24$ dB — about **12 dB more** than 2-ASK and **6 dB more** than 4-ASK.

The consistent **6 dB gap** between successive orders is a direct consequence of the fixed $d_{\min}$ constraint: each doubling of M roughly quadruples the average symbol energy, which translates to a 6 dB penalty in required SNR.

The simulated curves match the theoretical curves closely across all SNR values, confirming the correctness of the implementation.

3 Part III: Performance Comparison of BPSK, QPSK, 4QAM, and 4ASK

3.1 Overview

In this part, we compare the Bit Error Rate (BER) performance of three modulation schemes — rectangular BPSK, QPSK, and 4QAM — under the constraint of a fixed minimum distance $d_{\min} = 2$ between adjacent constellation points. The BER curve of 4-ASK (obtained in Part II) is also included on the same figure for reference.

3.2 Constellation Design and Minimum Distance Constraint

To ensure a fair comparison, all modulation schemes are designed so that the minimum Euclidean distance between any two adjacent symbols is held constant at $d_{\min} = 2$.

- **BPSK**: 1 bit per symbol. Constellation: $\{-1, +1\}$. Average symbol energy $E_s = 1$.
- **QPSK**: 2 bits per symbol. Rectangular constellation with independent binary signalling on the I and Q axes: $\{(\pm 1, \pm 1)\}$, Gray coded. Average symbol energy $E_s = 2$, hence $E_b = 1$.
- **4QAM**: 2 bits per symbol. Square QAM constellation: $\{(\pm 1, \pm 1)\}$, Gray coded. Identical to rectangular QPSK in terms of constellation geometry and energy; $E_s = 2$, $E_b = 1$.
- **4-ASK** (from Part II): 2 bits per symbol. Levels $\{-3, -1, +1, +3\}$, Gray coded. Average symbol energy $E_s = 5$, hence $E_b = 2.5$.

3.3 Detection Strategy

- **BPSK**: A single threshold at zero. A received sample $r \geq 0$ is decoded as bit 1; otherwise as bit 0.
- **QPSK / 4QAM**: Because the I and Q axes are orthogonal and statistically independent under AWGN, optimal detection reduces to two independent BPSK decisions — one threshold at zero on the I channel and one on the Q channel.
- **4-ASK**: Maximum-likelihood (nearest-neighbour) detection among the four levels $\{-3, -1, +1, +3\}$.

3.4 Results and Figure

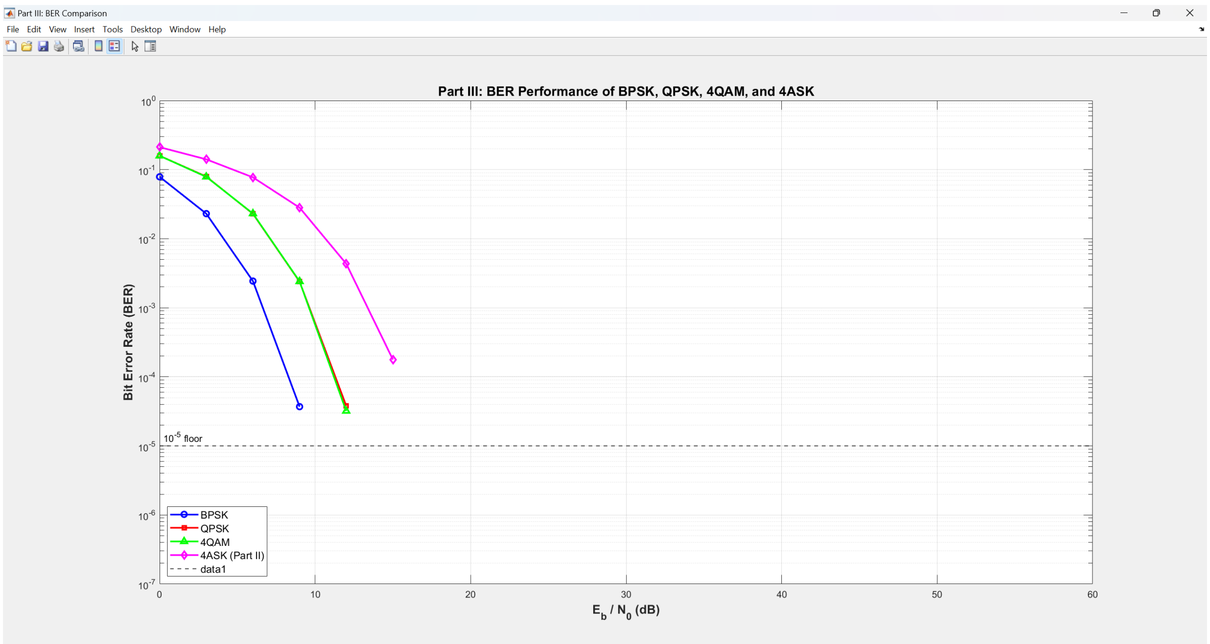

Figure 4: partIII:BER performance od BPSK, QBSK, 4QAM and 4ASK

3.5 comments

3.5.1 Which modulation scheme has the best performance, and why?

From Figure 4, **BPSK achieves the best BER performance**, closely followed by QPSK and 4QAM whose curves are nearly indistinguishable from BPSK. This result can be understood theoretically:

- For BPSK, the exact BER is

$$P_e^{BPSK} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \tag{1}$$

- For rectangular QPSK and 4QAM, the two orthogonal axes are independent BPSK channels. Each bit therefore experiences the same Q-function BER:

$$P_e^{QPSK} = P_e^{4QAM} = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \tag{2}$$

  The per-bit BER is thus *identical* to BPSK when $d_{\min}$ and E_b are equal. The simulation confirms this: all three curves overlap.

- For 4-ASK, the BER expression is

$$P_e^{4ASK} = \frac{3}{4}Q\left(\sqrt{\frac{2E_b}{5\,N_0/2}}\right) = \frac{3}{4}Q\left(\sqrt{\frac{4E_b}{5\,N_0}}\right) \tag{3}$$

  Because the four levels span a larger amplitude range, the *average* symbol energy is $E_s = 5$ (compared to $E_s = 2$ for QPSK/4QAM at the same $d_{\min}$). Consequently, 4-ASK requires significantly more E_b/N_0 to achieve the same BER, and its curve lies well to the right of the others.

In summary, QPSK and 4QAM achieve the same reliability as BPSK while doubling the spectral efficiency (bits per symbol), making them superior in bandwidth-limited systems. 4-ASK, despite carrying the same number of bits per symbol as QPSK/4QAM, pays a heavy penalty in energy efficiency due to the unequal distances of outer symbols from the origin.

3.5.2 At which SNR value does the system become nearly error-free?

A system is considered "nearly error-free" when BER $< 10^{-5}$.

Table 2: Approximate E_b/N_0 threshold for near-error-free operation (BER $< 10^{-5}$).

Modulation	**Required E_b/N_0 (dB)**
BPSK	≈ 9 dB
QPSK	≈ 9 dB
4QAM	≈ 9 dB
4-ASK	≈ 16 dB

From Figure 4 and Table 2:

- **BPSK, QPSK, and 4QAM** all reach BER $< 10^{-5}$ at approximately $E_b/N_0 \approx 9$ dB.
- **4-ASK** requires approximately $E_b/N_0 \approx 16$ dB — about 7 dB more than the other three — to achieve the same reliability level. This penalty arises from the higher average energy needed to maintain $d_{\min} = 2$ across four amplitude levels.